

$$\begin{aligned}
\text{(ii)} \quad \nabla(\mathbf{A} \cdot \mathbf{B}) &= \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\
\text{(iv)} \quad \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\
\text{(vi)} \quad \nabla \times (\mathbf{A} \times \mathbf{B}) &= (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})
\end{aligned}$$

(a) Check product rule (iv) for the functions

$$\mathbf{A} = x\hat{x} + 2y\hat{y} + 3z\hat{z}, \quad \mathbf{B} = 3y\hat{x} - 2x\hat{y}$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

Solution:

(a) Start with LHS:

$$\begin{aligned}
\nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \nabla \cdot ((A_y B_z - A_z B_y)\hat{x} + (A_z B_x - A_x B_z)\hat{y} + (A_x B_y - A_y B_x)\hat{z}) = \\
&= \frac{\partial}{\partial x}(A_y B_z - A_z B_y) + \frac{\partial}{\partial y}(A_z B_x - A_x B_z) + \frac{\partial}{\partial z}(A_x B_y - A_y B_x) = \\
&= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z = \boxed{15z}
\end{aligned}$$

RHS:

$$\begin{aligned}
&\underbrace{\mathbf{B} \cdot (\nabla \times \mathbf{A})}_{(1)} - \underbrace{\mathbf{A} \cdot (\nabla \times \mathbf{B})}_{(2)} = \\
(1) \quad &\mathbf{B} \cdot \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \right] = \\
&= B_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + B_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + B_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) = 0 \\
(2) \quad &\mathbf{A} \cdot \left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z} \right] = \\
&= A_x \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + A_y \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + A_z \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) = \\
&= 3z(-5) = -15z \\
&(1) - (2) = 0 - (-15z) = \boxed{15z}
\end{aligned}$$

(b) LHS:

$$\begin{aligned}
\nabla(\mathbf{A} \cdot \mathbf{B}) &= \nabla(A_x B_x + A_y B_y + A_z B_z) = \nabla(3xy - 4xy) = \\
&= \frac{\partial(-xy)}{\partial x} \hat{x} + \frac{\partial(-xy)}{\partial y} \hat{y} = \boxed{-y \hat{x} - x \hat{y}}
\end{aligned}$$

RHS:

$$\underbrace{\mathbf{A} \times (\nabla \times \mathbf{B})}_{(1)} + \underbrace{\mathbf{B} \times (\nabla \times \mathbf{A})}_{(2)} + \underbrace{(\mathbf{A} \cdot \nabla) \mathbf{B}}_{(3)} + \underbrace{(\mathbf{B} \cdot \nabla) \mathbf{A}}_{(4)}$$

$$(1) \mathbf{A} \times \left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z} \right] =$$

$$= (x\hat{x} + 2y\hat{y} + 3z\hat{z}) \times (-5\hat{z}) = -10y\hat{x} + 5x\hat{y}$$

$$(2) \mathbf{B} \times \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \right] = 0$$

$$(3) (\mathbf{A} \cdot \nabla) \mathbf{B} =$$

$$= \left(A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} \right) \hat{x}$$

$$+ \left(A_x \frac{\partial B_y}{\partial x} + A_y \frac{\partial B_y}{\partial y} + A_z \frac{\partial B_y}{\partial z} \right) \hat{y}$$

$$+ \left(A_x \frac{\partial B_z}{\partial x} + A_y \frac{\partial B_z}{\partial y} + A_z \frac{\partial B_z}{\partial z} \right) \hat{z} =$$

$$= 6y\hat{x} - 2x\hat{y}$$

$$(4) (\mathbf{B} \cdot \nabla) \mathbf{A} =$$

$$= \left(B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z} \right) \hat{x}$$

$$+ \left(B_x \frac{\partial A_y}{\partial x} + B_y \frac{\partial A_y}{\partial y} + B_z \frac{\partial A_y}{\partial z} \right) \hat{y}$$

$$+ \left(B_x \frac{\partial A_z}{\partial x} + B_y \frac{\partial A_z}{\partial y} + B_z \frac{\partial A_z}{\partial z} \right) \hat{z} =$$

$$= 3y\hat{x} - 4x\hat{y}$$

$$(1) + (2) + (3) + (4) = (-10y + 6y + 3y)\hat{x} + (5x - 2x - 4x)\hat{y} = -y\hat{x} - x\hat{y}$$

(c) LHS:

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \nabla \times [(A_y B_z - A_z B_y)\hat{x} + (A_z B_x - A_x B_z)\hat{y} + (A_x B_y - A_y B_x)\hat{z}] =$$

$$= \nabla \times [6xz\hat{x} + 9yz\hat{y} - (2x^2 + 6y^2)\hat{z}] = \nabla \times \mathbf{C} =$$

$$= \left(\frac{\partial C_z}{\partial y} - \frac{\partial C_y}{\partial z} \right) \hat{x} + \left(\frac{\partial C_x}{\partial z} - \frac{\partial C_z}{\partial x} \right) \hat{y} + \left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y} \right) \hat{z} =$$

$$= (-12y - 9y)\hat{x} + (6x + 4x)\hat{y} = -21y\hat{x} + 10x\hat{y}$$

RHS: From (b) it was calculated that $(\mathbf{A} \cdot \nabla) \mathbf{B} = 6y\hat{x} - 2x\hat{y}$ and $(\mathbf{B} \cdot \nabla) \mathbf{A} = 3y\hat{x} - 4x\hat{y}$. The remaining parts of (vi) RHS is

$$\underbrace{A(\nabla \cdot B)}_{=0} - \underbrace{B(\nabla \cdot A)}_{=6} = -6B = -18y\hat{x} + 12x\hat{y}$$

Summing it all up according to (vi), the result is

$$(3y - 6y - 18y)\hat{x} + (-4x + 2x + 12x)\hat{y} = -21y\hat{x} + 10x\hat{y}$$

□
